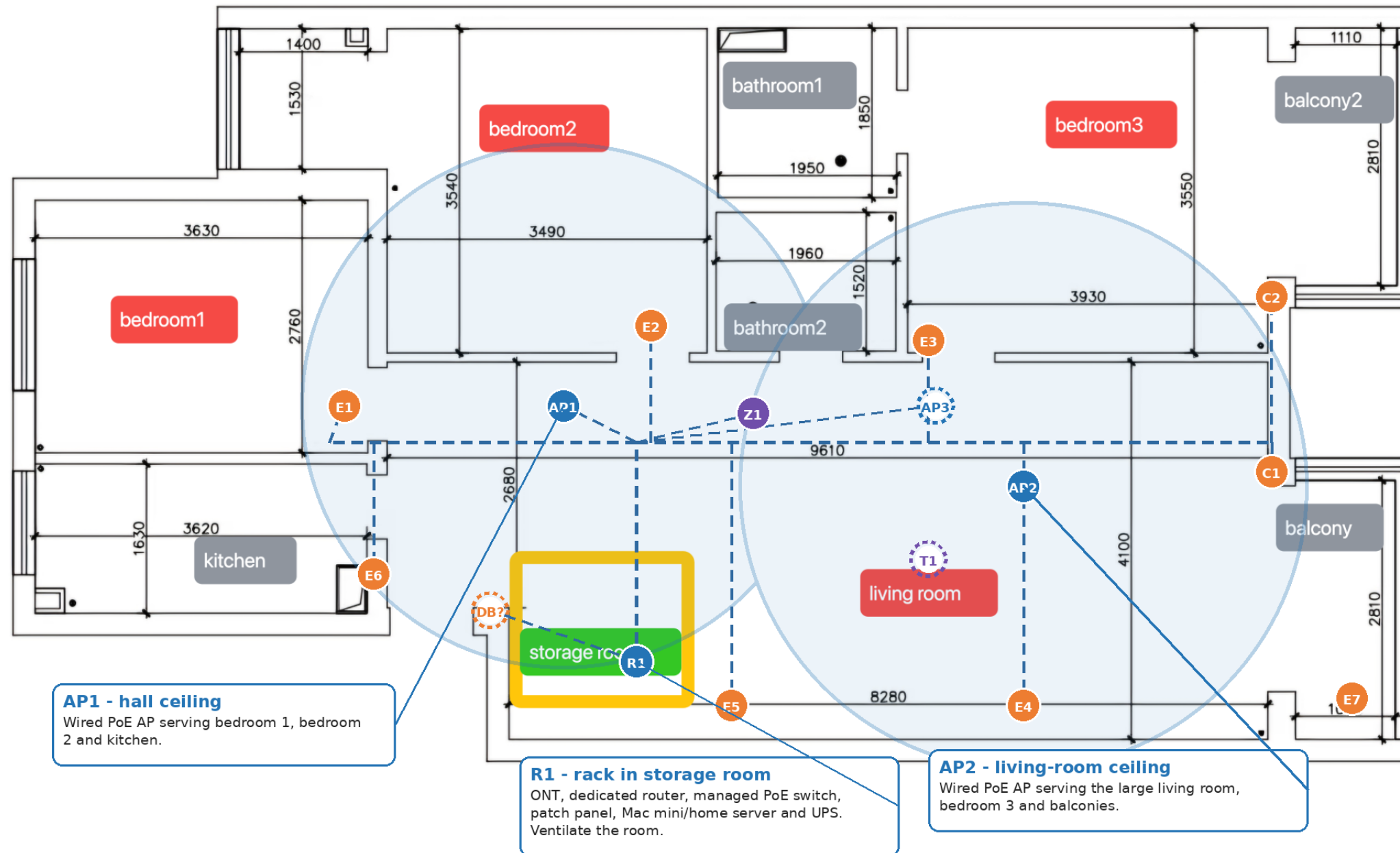


NETWORK & CABLING PLAN

Provisional layout based on the supplied floor plan



Core design

- Two active ceiling APs; one spare AP cable for later RF testing.
- Every Ethernet run is home-run to R1; no daisy-chaining.
- Keep Zigbee/Thread radios outside the metal rack.
- Bedroom 2 is provisionally treated as the office; move E2 to the actual office if different.
- The lower living-room wall is provisionally treated as the TV/media wall.

Marker schedule

- R1: 9U/12U ventilated rack.
- AP1/AP2: wired Wi-Fi access points.
- AP3: spare ceiling cable only.
- Z1: Ethernet/PoE Zigbee coordinator.
- T1: optional Thread border router.
- E1: bedroom 1 - 2 x Cat6.
- E2: bedroom 2/office - 4 x Cat6A.
- E3: bedroom 3 - 2 x Cat6.
- E4: TV/media wall - 4 x Cat6.
- E5: living-room auxiliary - 2 x Cat6.
- E6: kitchen cabinet - 1 x Cat6.
- DB?: PoE doorbell; entrance position must be confirmed.
- C1/C2: PoE camera points at balconies.
- E7: balcony/laundry spare - 1 x Cat6.

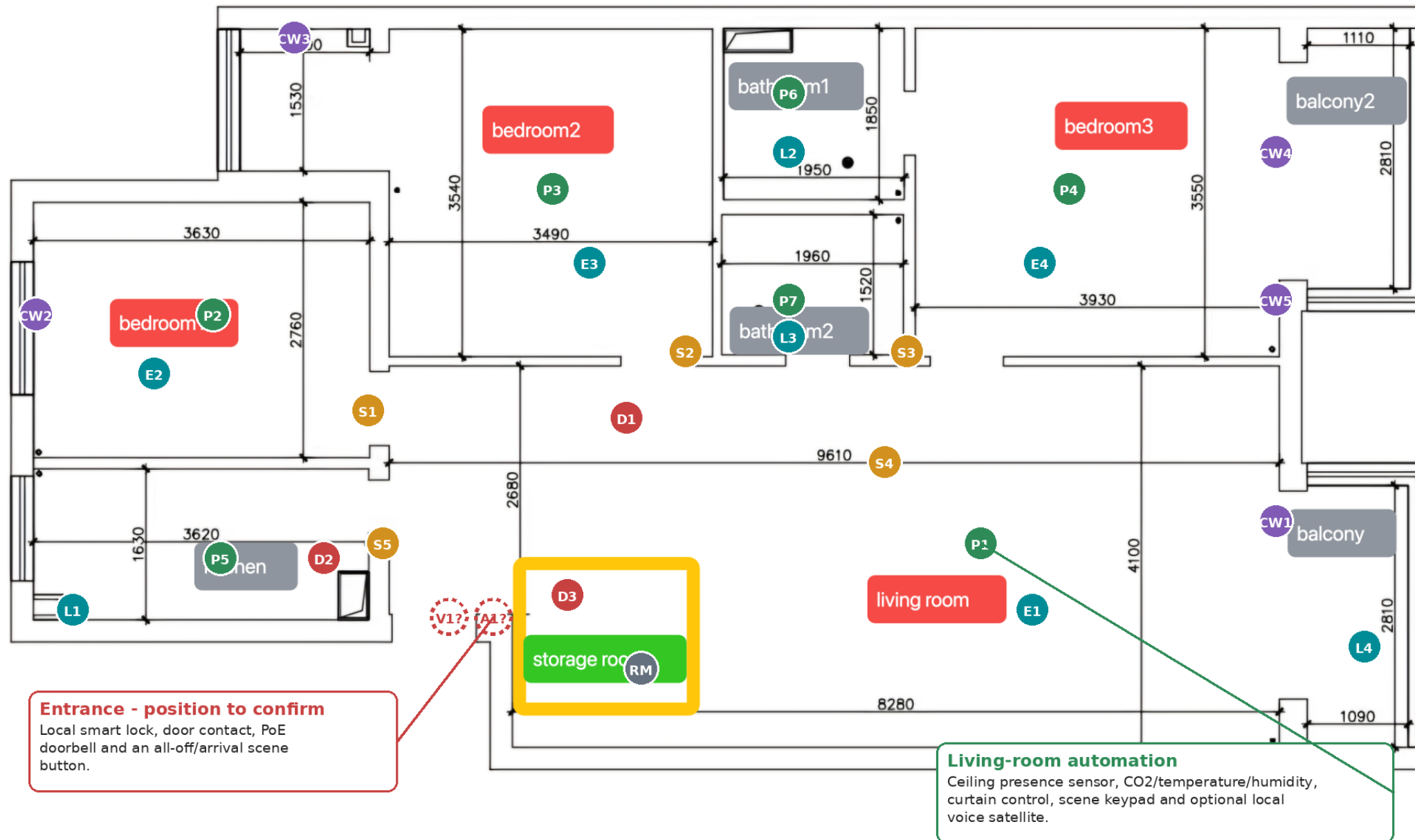
Rack & switch

- 24-port patch panel.
- 24-port managed PoE+ switch with 2-4 x 10G/SFP+ uplinks.
- At least four 2.5G ports for APs, server and workstation.
- 1000-1500 VA UPS for ONT, router, switch and home server.

Blue circles are illustrative only - final AP tuning requires an on-site RF survey.

SMART-HOME DEVICE PLAN

Local-first, self-managed and manually operable



Marker schedule

- P: ceiling presence/occupancy sensor.
- E: temperature/humidity; living E1 also measures CO2.
- CW: curtain-motor power plus door/window contact.
- L: water-leak sensor.
- S: local wall switch or scene keypad with neutral and physical operation.
- D1: certified hall smoke alarm with optional status integration.
- D2: certified kitchen heat/gas detection as locally required.
- D3: storage/rack smoke alarm.
- A1?: smart lock and entrance contact; confirm the actual main door.
- V1?: motorised main-water shutoff; confirm the incoming-water location.
- RM: rack temperature, UPS and ventilation monitoring.

Room package

- Each bedroom: one presence sensor, one temperature/humidity sensor, curtain power/contact and scene control.
- Living room: presence, CO2/T/RH, curtain contact, scene keypad and optional voice.
- Bathrooms: occupancy, humidity, leak sensor and local exhaust-fan relay.
- Kitchen: presence, under-sink leak sensor, certified detector and lighting control.
- Balcony/laundry: door contact, curtain/shade control and leak sensor near washer.

Renovation rules

- Neutral wire and deep box at every switch.
- Important lights, curtains and valves must work manually without Home Assistant.
- Use local-capable relays; avoid cloud-only devices.
- Safety alarms remain independent; automation adds notification only.

Network, server and software design

Assumptions: approximately 100 m²; 1 Gbit/s internet; storage room used as the rack room; bedroom 2 provisionally used as the office; lower living-room wall provisionally used as the TV/media wall; main entrance position inferred from the drawing and must be confirmed.

Item	Qty	Recommended specification / role
Dedicated router	1	Fanless x86 appliance, 4 x 2.5GbE preferred; OPNsense or OpenWrt; do not host the router on the home-automation server.
Managed PoE switch	1	24 RJ45 ports, VLAN support, 8-12 PoE+ ports minimum, at least four 2.5GbE ports, 2-4 10G/SFP+ uplinks.
Ceiling AP	2 + 1 cable	Two active wired APs; install one spare cable and add the third AP only after RF testing.
Rack	1	9U/12U ventilated rack, 24-port patch panel, cable management, PDU and environmental monitoring.
UPS	1	1000-1500 VA for ONT, router, switch, Mac mini/home server and radio gateways.
Home server	1	Use the 24/7 Mac mini initially. Run Home Assistant OS in a VM; keep custom applications in a separate VM/container environment.
Zigbee coordinator	1	Ethernet/PoE coordinator mounted in the central hall, outside the metal rack.
Thread border router	0-1	Add only when Thread devices are purchased; keep it on the automation LAN.

Cabling schedule

Location	Runs	Cable / purpose
AP1, AP2, AP3 reserve	3	Cat6 solid copper, PoE; AP ports preferably 2.5GbE.
Bedroom 1	2	General data; wire both to the rack.
Bedroom 2 / provisional office	4	Cat6A preferred; Dell workstation, dock/phone and future 10G path.
Bedroom 3	2	General data.
Living TV/media wall	4	TV, media player, console and spare.
Living auxiliary point	2	Flexible desk, AV or future access point.
Kitchen cabinet	1	Display, gateway or appliance integration.
Entrance	1	PoE doorbell; verify exact door position.
Balconies	3	Two PoE cameras plus one laundry/outdoor spare.
Zigbee/Thread positions	2	Ethernet/PoE for coordinator and optional border router.

Logical network

VLAN	Devices	Policy
10 Trusted	Phones, laptops, Dell, admin workstation	May reach approved local services and the internet.
20 Automation	Home Assistant, MQTT, Zigbee/Thread, ESPHome	Local-first; tightly limited internet access.
30 Untrusted IoT	TVs and cloud-dependent appliances	Internet allowed; blocked from trusted computers.
40 Cameras	PoE cameras and doorbell	May reach NVR, DNS and time only.
50 Guest	Visitors	Internet only.
99 Management	Router, switch and APs	Admin devices only.

Smart-home device schedule and implementation

Area	Base devices	Notes
Living room	1 presence sensor; 1 CO2/T/RH sensor; curtain/contact; scene keypad; optional voice satellite	Use wired power where possible. Keep voice and AI optional, not part of safety logic.
Each bedroom	1 presence/PIR; 1 T/RH sensor; window/door contact; curtain power; scene switch	Use battery Zigbee/Thread sensors and powered relays/curtain motors.
Kitchen	Presence; under-sink leak; certified heat/gas detector; lighting relay	Detector must remain independently functional.
Each bathroom	Occupancy; humidity; leak sensor; exhaust-fan relay	Fan automation should have local/manual override.
Entrance	Smart lock; door contact; PoE doorbell; arrival/all-off button	Exact entrance location must be confirmed.
Balconies	Door contacts; curtain/shade control; outdoor camera points; leak near washer	Select weather-rated devices where exposed.
Storage/rack	Temperature; smoke alarm; UPS state; fan control	Do not place Zigbee/Thread radios inside a metal cabinet.
Water	Motorised main shutoff valve plus leak sensors	Locate the valve at the actual incoming water line with manual override.

Recommended local software stack

Layer	Recommendation
Router/firewall	Dedicated OPNsense or OpenWrt appliance. Remote access through WireGuard; do not publish Home Assistant or cameras directly to the internet.
Core automation	Home Assistant OS in a VM on the 24/7 Mac mini. Configure automatic VM start after power restoration.
Message/device layer	Mosquitto MQTT, Zigbee2MQTT and ESPHome. Use an Ethernet Zigbee coordinator to avoid USB pass-through issues.
Custom applications	Separate Debian/Linux VM or container host for Python, Go, Node.js, Node-RED, databases and dashboards.
Heavy compute	Use the Dell RTX 3090 workstation as an optional worker for local AI, speech or video analysis. Core lights and safety logic must continue when it is off.
Backups	Nightly Home Assistant backup plus configuration repository; copy encrypted backups to separate storage.

Commissioning sequence

1	Before walls close	Confirm furniture/TV/office/entrance, install conduit, neutral wires, deep boxes, curtain power, AP/PoE/data runs and spare pull strings.
2	Cable verification	Label both ends and test every copper run. Photograph concealed routes.
3	Network	Install router, VLANs, switch and APs; measure 5 GHz coverage before buying AP3.
4	Core automation	Bring up Home Assistant, MQTT, Zigbee2MQTT and ESPHome; configure local backups and WireGuard.
5	Room devices	Add powered Zigbee routers first, then battery sensors, curtains, relays and leak protection.
6	Safety tests	Verify manual operation, internet-out operation, server-down operation, UPS runtime and water-shutoff behaviour.